Week 4 — Technical Architecture & Constraints

> *"A TPM doesn't choose the architecture. A TPM names the constraints — clearly, in business terms — that the architecture must respect."*

Week 4 is the technical-thinking week. Each triad takes the **PRD they shipped Friday of Week 3** and treats it as the input to an architectural conversation: monolith vs microservices, security & compliance, system components, latency / availability / rate limits, and the trade-offs that fall out of those decisions.

By Friday, every triad produces a **Technical Concept Document (TCD)** — a one- to two-page artifact that an architect or engineering lead would accept as scoping input. The TCD is a sibling to the PRD; together they go into Week 5.

What's different about Week 4

- **AI is restored.** After last week's discipline, AI returns as a research and structuring assistant. The Week 1 / Week 2 prompt patterns are reused; the Week 3 review discipline applies to AI-assisted output.
- **You are not pretending to be an engineer.** The job is to make trade-offs visible and defensible to non-technical stakeholders. The architect makes the call; the TPM ensures the call is made on the right inputs.
- **The PRD's NFR section is a first draft.** Every architectural decision this week tightens, rewrites, or relocates an NFR.

Learning outcomes

By Friday afternoon, each participant can:

1. Frame a **monolith-vs-microservices** decision in business and operational terms — not in framework hype.
2. Conduct an architecture-level **STRIDE threat-model pass** and translate the results into PRD security NFRs an engineer can scope against.
3. Draw a **C4-style component diagram** (Context + Container) for a feature, naming external dependencies and integration points.
4. Set realistic **SLO targets** (latency, availability, rate limits) anchored to user behavior and business cost, with an **error budget** the team will actually respect.

5. Author a **Technical Concept Document** that surfaces the top five constraints, names trade-offs explicitly, and ties each to a stakeholder who must approve.

Daily map

Day	Topic	Key artifact produced
1	Monolith vs Microservices	Architecture stance + integration choice
2	System security & compliance	STRIDE threat model + revised security NFRs
3	Mapping high-level system components	C4 Context + Container diagram
4	Latency, availability, rate limits	SLO sheet + latency budget + rate-limit policy
5	Technical trade-offs and constraints	**Technical Concept Document (TCD)**

Each day's output is a section of the TCD. Friday assembles, peer-reviews, and ships the TCD alongside the PRD.

Daily cadence (applies all five days)

Clock	Block	Mix
09:00 – 09:15	Opening & objectives	Instructor
09:15 – 10:30	Teaching Block 1 + Activity 1	~25 min teach / 50 min triad work
10:30 – 10:45	**Break**	
10:45 – 12:00	Teaching Block 2 + Activity 2	~25 min teach / 50 min triad work
12:00 – 13:00	**Lunch**	
13:00 – 14:15	Teaching Block 3 + Activity 3	~25 min teach / 50 min triad work
14:15 – 14:30	**Break**	
14:30 – 16:00	Teaching Block 4 + Activity 4 + Wrap	~25 min teach / 45 min triad / 20 min wrap

The TCD template (Friday's deliverable)

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# Technical Concept Document — <feature name>
**Sibling to:** PRD <link>  |  **Authors:** <triad>  |  **Status:**
Draft / Reviewed

## 1. Architecture stance
Monolith / microservice / hybrid; one paragraph defending the call from
business + operational angles (not framework hype).

## 2. Integration map
Which existing systems / services this feature depends on, calls, or
extends.
References the C4 Container diagram (Day 3 artifact).

## 3. Security & compliance constraints
STRIDE pass summary; the 3–5 highest-priority threats and their
mitigations.
Updated security and compliance NFRs (replaces the Week-3 first draft).

## 4. SLO + rate-limit policy
- Latency: p50 / p95 / p99 targets, with defenses
- Availability: monthly target + error budget
- Rate limits: per-user / per-tenant / global; what each protects

## 5. Top 5 trade-offs
Each trade-off names: option A, option B, the call we made, the cost we
accept,
the trigger that would cause us to revisit.

## 6. Stakeholders + sign-off matrix
| Constraint | Stakeholder | Status (proposed / discussed / approved) |
```

Triads

Same triads from Weeks 1–3. They authored the PRD; they author the TCD.

Friday review rubric (TCD)

Dimension	Weight	What "exemplary" looks like
Stance defensibility	20%	Architecture call defended in business terms; framework hype absent
Threat-model rigor	20%	3–5 named threats with concrete mitigations; not boilerplate
Component clarity	15%	C4 diagram is readable; external dependencies named
SLO realism	20%	Targets anchored to user behavior; error budget the team will actually keep
Trade-off honesty	15%	Both sides named; trigger to revisit specified
Stakeholder map	10%	Each constraint has an owner; sign-off status is honest

Bridge to Week 5

Week 5 (Technical Infrastructure & Modeling) goes one level deeper: databases, cloud architecture, REST/SOAP APIs, performance baselines, monitoring. The **TCD is the input** — its component map drives Week 5's data-modeling work; its SLOs drive performance-baseline conversations; its threat model drives encryption and key-management decisions.